

Nmap Network Scanning Lab (Beginner)

Objective

The goal of this lab was to learn how to safely scan a local network to see what devices are present, what network services are running, and what software is being used. This lab was completed in an isolated virtual environment to avoid scanning real or restricted networks.

Lab Setup

Hypervisor: VMware Workstation

Operating System: Ubuntu Linux

Lab-01: Scanner machine

Lab-02: Target machine

Network Type: Host-Only (isolated from the internet)

Tools Used

Nmap – network scanning tool

OpenSSH Server – service used to expose a network port for scanning

Steps Performed

1. Installed Nmap on Lab-01 using NAT networking.
2. Switched both virtual machines to Host-Only networking for safe scanning.
3. Identified IP addresses of both machines.
4. Scanned the network to discover active hosts.
5. Installed and enabled SSH on Lab-02.
6. Re-scanned Lab-02 from Lab-01 using Nmap.
7. Identified open ports and services.
8. Detected the SSH service version.

Commands Used

```
nmap -sn 192.168.174.0/24
```

```
nmap -Pn 192.168.174.129
```

```
nmap -sV -Pn 192.168.174.129
```

Results

Lab-02 was reachable on the network.

Port 22 (SSH) was open.

The SSH service was identified as OpenSSH.

No other services were exposed.

Conclusion

This lab demonstrated how network scans can identify exposed services and software running on a system. It showed the importance of limiting exposed services to reduce security risk.

Key Takeaway

If a computer is connected to a network, tools like Nmap can reveal what services it is offering and what software is running.